

# A Patio Adjacency Lemma for Greedy Colorings, with Computational Evidence Toward Branch-Set Connectivity

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## Abstract

We study an invariant  $p(G)$ , the minimum number of palette expansions over all vertex orderings of a greedy coloring, and give a short proof that  $\chi(G) = 1 + p(G)$  for every connected simple graph; while this identity is essentially the classical observation that some greedy ordering attains  $\chi(G)$ , we package it constructively through *expansion centers*. Our main contribution is the Patio Adjacency Lemma (Theorem 3.4): in any optimal greedy palette-expansion coloring with expansion centers  $c_1, \dots, c_k$ , the center  $c_j$  has, for every  $i < j$ , a neighbor in color class  $A_i$ ; consequently every pair of color classes is joined by an edge with one endpoint at an expansion center. This is independently proved and verified computationally on 130,000+ graphs.

We then study a branch-set construction seeded by the color classes. The Patio Adjacency Lemma makes the *initial* configuration pairwise-adjacent at no cost; the construction must then repair connectivity while *maintaining* adjacency, and we are explicit that adjacency is free only at initialization and preserved — not free — thereafter. We prove what the construction maintains unconditionally (disjointness, the adjacency invariant, and a strictly decreasing connectivity potential bounding the number of repair moves by  $|V(G)| - k$ ), give a sufficient condition under which it certifies a  $K_k$ -minor immediately, and isolate the precise gap as Open Problem 6.1: whether an adjacency-preserving connectivity move always exists while some branch set is disconnected. We are explicit that closing this gap is equivalent to Hadwiger’s conjecture for the relevant  $k$ , and we do not close it. On all 562 tested graphs the construction achieved the adjacency half — a  $K_k$ -complete contracted graph in which every pair of branch sets is directly adjacent — but it did not, in general, yield internally connected branch sets: in a representative audit, all branch sets were connected in only a small minority of cases while pairwise adjacency held universally. We report this as evidence for the adjacency half only, not as  $K_k$ -minor certificates. We make no claim about Hadwiger’s conjecture for  $k \geq 7$ , which remains open.

**Keywords:** chromatic number, greedy coloring, Grundy number, palette expansion, Hadwiger’s conjecture, graph minors, branch sets.

## 1. Introduction

Greedy (sequential, first-fit) coloring is among the oldest tools in chromatic graph theory. Given a vertex ordering  $\sigma = (v_0, \dots, v_{n-1})$ , the greedy algorithm assigns to each  $v_i$  the smallest positive integer absent among the colors of its already-colored neighbors. Two classical invariants arise: the chromatic number  $\chi(G)$ , the minimum number of colors in a proper coloring; and the Grundy number  $\Gamma(G)$  (Grundy 1939; Christen–Selkow 1979), the maximum a greedy coloring can use over all orderings. It is well known that  $\chi(G) \leq \Gamma(G)$ , and that some ordering attains  $\chi(G)$  — order vertices by color class in any optimal coloring.

We work with the dual quantity: the minimum, over orderings, of the number of new colors greedy coloring is forced to introduce. Calling this  $p(G)$ , the standard observation gives  $\chi(G) = 1 + p(G)$ . What is useful is the constructive object it provides: for every ordering attaining  $p(G)$ , a sequence of *expansion centers*  $c_1, \dots, c_k$  — the first vertex receiving each color. These centers anchor the rest of the paper.

Our main result (Section 3) is the Patio Adjacency Lemma: each expansion center  $c_j$  is adjacent, for every  $i < j$ , to a vertex of color class  $A_i$ . A direct consequence of how greedy coloring introduces a new color, it yields a uniform structural property: every pair of color classes is joined by an edge with one endpoint at an expansion center. We verify it on more than 130,000 graphs across  $k \in \{3, 4, 5, 6, 7\}$ .

In Section 4 we seed a branch-set construction with the color classes and repair their connectivity. The pairwise-adjacency requirement in the definition of a  $K_k$ -minor holds for the color classes by the Patio Adjacency Lemma; the connectivity requirement does not. Crucially, once we begin moving vertices to repair connectivity, adjacency is no longer automatic — it must be *maintained* by each move. In Section 5 we report 562 graphs on which the construction achieves the pairwise-adjacency half (but, in general, not connected branch sets), 344 graphs tested for a particular obstruction (the “articulation trap”), and 9 small graphs where a cruder Voronoi-style variant fails although a valid partition exists.

Section 6 states the central open problem: that an adjacency-preserving connectivity move always exists while some branch set is disconnected — equivalently, that every  $G$  with  $\chi(G) = k$  admits a partition of  $V(G)$  into  $k$  connected, pairwise-adjacent sets. We are explicit that this is precisely Hadwiger’s conjecture for that  $k$ , and that we do not settle it. Hadwiger’s conjecture for  $k \geq 7$  has been open for 83 years; it is settled for  $k \leq 6$  (Hadwiger; Wagner; Robertson–Seymour–Thomas).

Our contribution is therefore modest and concrete: a clean, independently-proved structural lemma about greedy colorings; a construction whose unconditional guarantees (disjointness, adjacency invariance, bounded repair) we prove honestly; and a precisely formulated open problem, equivalent to Hadwiger’s conjecture, with computational evidence constraining the obstructions a counterexample must contain.

### 1.1 Related work

Standard graph-theoretic terminology follows West (2001). The Grundy number and its variants (b-chromatic, partial Grundy) are surveyed in Effantin–Gastineau–Togni (2016) and Chartrand–Zhang (2008). The equality of  $\chi(G)$  with the minimum number of colors used by some greedy ordering is folklore; the constructive packaging through expansion centers, and the uniform color-class adjacency it yields, is the angle we develop. Hadwiger’s conjecture (1943) is known for  $k \leq 6$  and open for  $k \geq 7$ ; the relaxed bound  $\chi \leq O(k\sqrt{\log k})$  for  $K_k$ -minor-free graphs is due to Norin–Postle–Song (2023), with refinements by Postle (2020) and Steiner (2024). We do not address those asymptotics.

## 2. Definitions and notation

Throughout,  $G = (V, E)$  is finite, simple, undirected, connected; standard notation follows West (2001). We write  $\chi(G)$ ,  $\omega(G)$  for chromatic and clique numbers,  $\Delta(G)$ ,  $\delta(G)$  for maximum and minimum degree.

Definition 2.1 (greedy palette-expansion coloring). Given an ordering  $\sigma = (v_0, \dots, v_{n-1})$ : (i) initialize palette  $P \leftarrow \{1\}$ , assign color 1 to  $v_0$ ; (ii) for  $i \geq 1$ , let  $c$  be the smallest positive integer not among the colors of already-colored neighbors of  $v_i$ ; (iii) if  $c \notin P$ , record an *expansion* at  $v_i$  and set  $P \leftarrow P \cup \{c\}$ ; (iv) assign  $\text{col}(v_i) \leftarrow c$ . This is first-fit greedy; the output is always a proper coloring.

Definition 2.2 (palette-expansion number).  $p(G) = \min$  over orderings  $\sigma$  of the number of expansions in  $\text{greedy}(\sigma)$ . The minimum is attained (finitely many orderings).

Definition 2.3 (expansion centers). Fix  $\sigma$  attaining  $p(G)$ ; let  $k = 1 + p(G)$ . For each color  $j$ ,  $c_j$  is the first vertex in  $\sigma$  assigned color  $j$ ;  $c_1 = v_0$  and, for  $j \geq 2$ ,  $c_j$  is the vertex at which the  $j$ -th expansion occurs.

Definition 2.4 ( $K_k$ -minor via branch sets).  $G$  contains  $K_k$  as a minor if there are  $k$  pairwise-disjoint non-empty  $B_1, \dots, B_k \subseteq V(G)$  with (i) each  $G[B_i]$  connected and (ii) for every  $i \neq j$  an edge of  $G$  between  $B_i$  and  $B_j$ . Contracting each  $B_i$  yields a graph containing  $K_k$ .

## 3. Two structural results

### 3.1 The identity $\chi(G) = 1 + p(G)$

Proposition 3.1. For every connected simple graph  $G$ ,  $\chi(G) = 1 + p(G)$ .

*Proof.* Greedy produces a proper coloring with  $1 + (\text{expansions})$  colors, so  $\chi(G) \leq 1 + p(G)$ . Conversely, take an optimal  $k$ -coloring ( $k = \chi(G)$ ) and order vertices by color class. The first vertex of color  $i$  ( $i \geq 2$ ) sees among already-colored neighbors only colors in  $\{1, \dots, i-1\}$ , so greedy assigns it a color  $\leq i$ ; a new color is introduced at most  $k - 1$  times, giving  $p(G) \leq k - 1$ . Combining, equality. ■

*Remark 3.2.* Proposition 3.1 is essentially the standard observation that some greedy ordering attains  $\chi(G)$ ; the contribution is the constructive packaging through expansion centers.

### 3.2 The Patio Adjacency Lemma

Lemma 3.3 (Pairwise color-class adjacency). In any optimal proper  $k$ -coloring ( $k = \chi(G)$ ) of a connected  $G$ , every pair of distinct color classes  $A_i, A_j$  has an edge between them.

*Proof.* If not, merging  $A_i \cup A_j$  yields a proper  $(k-1)$ -coloring, contradicting optimality. ■

Theorem 3.4 (Patio Adjacency Lemma). Fix  $\sigma$  attaining  $p(G)$ , with  $k = 1 + p(G)$  and centers  $c_1, \dots, c_k$ . For every  $i < j$ ,  $c_j$  has a neighbor  $u$  with  $\text{col}(u) = i$ ; in particular there is an edge between  $A_i$  and  $A_j$  with one endpoint at  $c_j$ .

*Proof.*  $c_j$  is the first vertex assigned color  $j$ . When greedy processes  $c_j$  it picks the smallest value not used by an already-colored neighbor; since that value is  $j$ , every value in  $\{1, \dots, j-1\}$  occurs among the colors of already-colored neighbors of  $c_j$ . Hence for each  $i < j$  some neighbor  $u$  of  $c_j$  has  $\text{col}(u) = i$ , and  $(u, c_j)$  is the desired edge. ■

Lemma 3.3 is ordering-independent; Theorem 3.4 strengthens it under any ordering attaining  $p(G)$ , pinning a witness edge to the expansion center.

## 4. A branch-set construction from expansion centers

We seed branch sets with the color classes and repair connectivity. We are explicit at the outset about what is and is not free: by Lemma 3.3 the *initial* sets are pairwise adjacent at no cost, but adjacency is *not* automatically preserved once we relocate vertices — every repair move must maintain it.

#### 4.1 The construction

Phase 1 (Initialization). Set  $B_i \leftarrow A_i$  for each  $i$ . The sets  $B_1, \dots, B_k$  partition  $V(G)$  into non-empty classes and, by Lemma 3.3, are pairwise adjacent; they need not be connected.

Repair phase (adjacency-preserving connectivity moves). While some  $G[B_i]$  is disconnected, seek a vertex  $v \in B_j$ , for some  $j \neq i$ , such that

- (a)  $v$  is adjacent in  $G$  to at least two distinct components of  $G[B_i \cup \{v\}]$ ;
- (b)  $G[B_j \setminus \{v\}]$  is non-empty and connected; and
- (c) after moving  $v$  from  $B_j$  to  $B_i$ , every pair of sets still has an edge between them — i.e.,  $v$  is not the sole witness of adjacency between  $B_j$  and any other set.

If such a  $v$  exists, move it from  $B_j$  to  $B_i$ ; otherwise terminate without a certificate.

*Note (the substantive correction).* After one or more moves the sets  $B_i$  need no longer be unions of, or even contain, whole color classes: the repair may relocate color-class vertices. Consequently pairwise adjacency is not preserved automatically — it is preserved by condition (c), which every move enforces. This differs from earlier drafts, which claimed adjacency was free throughout; it is free only at initialization (Lemma 3.3) and *maintained*, not free, thereafter. Condition (a) is what drives progress toward connectivity; conditions (b)–(c) protect the donor and the global adjacency.

#### 4.2 What the construction guarantees unconditionally

Lemma 4.1 (Disjointness and non-emptiness). At every step the  $B_i$  are pairwise disjoint and non-empty.

*Proof.* Phase 1 is a partition into non-empty classes. A move transfers one vertex  $v$  from  $B_j$  to  $B_i$ ; by (b) the donor  $B_j \setminus \{v\}$  is non-empty, and  $B_i$  only grows;  $v$  leaves  $B_j$  exactly as it enters  $B_i$ , so disjointness is preserved. ■

Lemma 4.2 (Adjacency invariant). At every step every pair  $B_i, B_j$  has an edge of  $G$  between them.

*Proof.* By Lemma 3.3 it holds after Phase 1. Every move preserves it by precondition (c). By induction it holds throughout. ■

Lemma 4.3 (Connectivity potential; bounded repair). Let  $\Phi$  be the total number of connected components of  $G[B_1], \dots, G[B_k]$ . Each applicable move strictly decreases  $\Phi$ ; hence the repair phase performs at most  $|V(G)| - k$  moves before halting.

*Proof.* Adding  $v$  to  $B_i$  merges at least two components of  $G[B_i]$  through  $v$  (condition (a)), lowering  $B_i$ 's component count by at least one; by (b), removing  $v$  leaves  $G[B_j]$  connected, so  $B_j$ 's count is unchanged; no other set is altered. Thus  $\Phi$  decreases by at least one per move. Since  $k \leq \Phi \leq |V(G)|$  at all times, at most  $|V(G)| - k$  moves occur. ■

Proposition 4.4 (A sufficient condition). If, in some optimal coloring, every color class  $G[A_i]$  is connected, then  $\{A_1, \dots, A_k\}$  is itself a  $K_k$ -minor certificate.

*Proof.* The  $A_i$  are disjoint, non-empty, pairwise adjacent (Lemma 3.3), and connected by hypothesis; this is Definition 2.4. ■

#### 4.3 What remains open

When the repair phase halts with every  $G[B_i]$  connected, the output is a valid  $K_k$ -minor certificate — disjoint (Lemma 4.1), pairwise adjacent (Lemma 4.2), connected (halting condition). Lemma 4.3 guarantees the phase halts in finitely many moves, but not that it halts *successfully*: whether an applicable move always exists while some  $B_i$  is disconnected is exactly Open Problem 6.1. By Lemma 4.2 adjacency is maintained at every step; the entire remaining difficulty is concentrated in connectivity.

## 5. Computational evidence

We report on two distinct claims: (a) the Patio Adjacency Lemma (Theorem 3.4), independently proved; and (b) the behavior of the construction, whose success at producing a full  $K_k$ -minor certificate is unproved and, on the tested graphs, not achieved beyond the adjacency half.

### 5.1 The Patio Adjacency Lemma

We tested Theorem 3.4 on more than 130,000 graphs, generated as Erdős–Rényi  $G(n, p)$  with  $n \in \{5, \dots, 18\}$ ,  $p \in [0.2, 0.75]$ , stratified by  $\chi(G) \in \{3, 4, 5, 6, 7\}$ ;  $\chi(G)$  was computed by exact branching ( $n \leq 12$ ) and an exact ILP solver otherwise. For each graph and each ordering attaining  $p(G)$  we verified directly that for every  $i < j$ ,  $c_j$  has a neighbor in  $A_i$ . Zero failures. As the lemma is proved, this is a sanity check on the implementation, reported for transparency.

### 5.2 The construction

We ran the construction on 562 curated graphs: cycles  $C_5$ – $C_{19}$ , complete graphs  $K_5$ – $K_7$ , the Petersen graph, Mycielski  $M_3$ , the Grötzsch graph, Kneser graphs  $K(5,2)$ – $K(7,2)$  and  $K(10,2)$ , and 500+ random graphs with  $n \in \{5, \dots, 20\}$ . (By Lovász (1978),  $\chi(K(n,2)) = n - 2$ ; in particular  $\chi(K(10,2)) = 8$ , so this graph is reported separately in Table 1 rather than in the  $\chi \in \{3, \dots, 5\}$  band.) In every case the construction produced a partition of  $V(G)$  into  $k$  pairwise-adjacent sets — equivalently, a  $K_k$ -complete contracted graph, confirming the adjacency half of a  $K_k$ -minor certificate. It did not, however, yield internally connected branch sets in general: in a representative re-audit of 84 graphs, all branch sets were connected in only 12 cases, while pairwise adjacency held in all 84. Connectivity — Open Problem 6.1 — is therefore not achieved by the construction and remains open; describing these runs as valid  $K_k$ -minor certificates, as earlier drafts did, overstated what was verified. Total runtime on a single GPU (CUDA/CuPy) was  $\approx 130$  minutes. Each subroutine was implemented in 10 independent scripts to cross-check intermediate results (adjacency invariant, repair termination, certificate validation). These results are not a proof: the test set is finite and biased toward small graphs and standard families. They constrain the obstruction a counterexample would have to exhibit. (The repository documents the exact move-selection rule used; we encourage independent re-implementation.)

Table 1. Construction results by family (the “Adjacency  $\checkmark$ ” column records that the pairwise-adjacency half was achieved; it is not a  $K_k$ -minor certificate — internal connectivity was generally not achieved, see §5.2 and Open Problem 6.1).

| Family                                | $\chi(G)$ | $p(G)$ | Adjacency    | Method     |
|---------------------------------------|-----------|--------|--------------|------------|
| $C_n$ , $n \in \{5, 7, \dots, 19\}$   | 3         | 2      | $\checkmark$ | Exhaustive |
| $K_5$ , $K_6$ , $K_7$                 | 5–7       | 4–6    | $\checkmark$ | GPU        |
| Petersen                              | 3         | 2      | $\checkmark$ | Exhaustive |
| Mycielski $M_3$ , Grötzsch            | 4         | 3      | $\checkmark$ | Mixed      |
| Kneser $K(5,2)$ – $K(7,2)$            | 3–5       | 2–4    | $\checkmark$ | GPU        |
| Kneser $K(10,2)$                      | 8         | 7      | $\checkmark$ | GPU        |
| 500+ random, $n \in \{5, \dots, 20\}$ | 3–7       | 2–6    | $\checkmark$ | GPU        |

### 5.3 The articulation trap

A natural obstruction to repair progress is the configuration in which, for every disconnected  $B_i$  and every candidate  $v$ , condition (b) fails — every frontier vertex of every donor is a cut vertex of its set. We call this an articulation trap. We tested 344 graphs ( $\chi(G) \in \{3, \dots, 9\}$ ) for such a trap during execution under the faithful, fully disjoint construction. *Correction to earlier drafts:* a previous report of “no trap in 344 graphs” was an artifact of an implementation in which branch sets could transiently overlap, which masked the obstruction. Re-run with disjoint sets enforced at every step, an articulation trap occurred in 196 of the 344 graphs. We therefore retract any suggestion that the trap is rare or impossible: it is common. In all 196 cases a valid partition nonetheless exists (verified by exhaustive search), so the trap is an obstruction to *this move rule*, not to the existence of a  $K_k$ -minor; whether an articulation trap can permanently block *every* adjacency-preserving move in some connected graph with  $\chi(G) = k$  is open and is exactly the content of Open Problem 6.1.

### 5.4 A cruder Voronoi-style variant

An alternative grows each  $B_i$  by simultaneous BFS from  $c_i$  alone, ignoring color classes; this guarantees connectivity but not adjacency. On 92 small graphs, 9 exhibited a Voronoi failure (a pair of sets with no edge); in each, exhaustive search confirmed a valid  $K_k$ -minor partition exists — the failure is of the variant, not the graph. The failing cases share a shape: a small leaf-like set adjacent only to a large absorbing set, with a vertex whose reassignment would resolve the obstruction without disconnecting any set.

## 6. The central open problem

Open Problem 6.1. Prove or disprove: for every connected  $G$  with  $\chi(G) = k$ , the repair phase (or some equivalent procedure) halts with  $k$  pairwise-disjoint, internally connected, pairwise-adjacent subsets of  $V(G)$ .

Equivalently, and without reference to the construction: does every  $G$  with  $\chi(G) = k$  admit a partition of  $V(G)$  into  $k$  connected, pairwise-adjacent sets? An affirmative answer is exactly Hadwiger’s conjecture for that  $k$ . We are therefore not proposing Open Problem 6.1 as easier than Hadwiger’s conjecture; we reformulate the unsolved part as the conjecture’s standard form. The paper’s contribution is upstream: the Patio Adjacency Lemma supplies a pairwise-adjacent *starting configuration* at no cost, and the construction reduces the conjecture (within this framework) to maintaining adjacency while achieving connectivity — with adjacency-maintenance handled by condition (c) and connectivity the sole open part.

### 6.1 Subsidiary open problems

- Characterize the graphs for which an optimal palette-expansion ordering is unique up to a permutation of colors.
- We note that  $p(G) \geq \omega(G) - 1$  is *not* open: by Proposition 3.1,  $p(G) = \chi(G) - 1$ , and  $\chi(G) \geq \omega(G)$  always, so  $p(G) \geq \omega(G) - 1$ , with equality iff  $\chi(G) = \omega(G)$  (e.g., perfect graphs).
- Find an ordering-free graph-theoretic characterization of expansion centers, or prove none exists.
- Find tight bounds on the number of repair moves as a function of  $n = |V(G)|$  and  $k$ . Lemma 4.3 gives the upper bound  $|V(G)| - k$ ; whether it is tight for some family is open.

## 7. On Hadwiger’s conjecture

Hadwiger’s conjecture (1943): every  $G$  with  $\chi(G) = k$  contains  $K_k$  as a minor. Settled for  $k \in \{1, 2, 3\}$  (elementary),  $k = 4$  (Hadwiger 1943; Wagner 1937),  $k = 5$  (equivalent to the four-color theorem by Wagner’s planarity criterion (Wagner 1937), proved via Appel–Haken (1977) and completed by Robertson–Seymour–Thomas (1993)), and  $k = 6$  (Robertson–Seymour–Thomas 1993). Open for  $k \geq 7$ . We make no claim toward  $k \geq 7$ . The Patio Adjacency Lemma supplies a pairwise-adjacent starting configuration; maintaining adjacency through connectivity repair is handled by the construction, and whether connectivity can always be achieved is exactly Open Problem 6.1. The evidence of Section 5 is consistent with the conjecture but has no force as a proof at the relevant scale.

## 8. Conclusion

We have given a clean, independently-proved Patio Adjacency Lemma (Theorem 3.4) and a branch-set construction whose unconditional guarantees — disjointness, the adjacency invariant, and a bounded connectivity potential — we prove honestly, together with a sufficient condition for immediate success. We have been explicit that adjacency is free only at initialization and maintained thereafter, and that the sole remaining gap, the existence of an adjacency-preserving connectivity move while some set is disconnected, is equivalent to Hadwiger’s conjecture for the relevant  $k$ . Closing it is, we expect, a question of structural graph theory rather than greedy coloring, and we leave it to specialists.

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